

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 03, 2026

VIDEO INFORMATION

Video ID:	0ecac639-6e57-4bab-8281-9d6feebe9599
Total Frames:	731
Frame Rate:	25.0 fps (assumed)
Duration:	29.2 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	626	✓
RT-DETR Detections	678	✓
MedSAM2 Detections	752	✓
Consensus (3-Model Agreement)	1	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	1	100.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	1	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — COLITIS [MEDIUM_RISK] (Tracked across 523 frames)



Feature	Value	Interpretation
Frame Index	414	Frame 414 of 731
Detection Method	Detector Consensus (MedSAM2 + U-Net)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	COLITIS	Type confidence: 90.7%
Redness Score	0.246	Color-based vascularization
Relative Radius	28.8%	Polyp size as % of frame diagonal
Texture Score	0.033	Surface morphology
Vessel Visibility	0.162	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket

Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	86.5%	YOLO / RT-DETR / track confidence

Type Assessment: Inflammatory mucosal pattern — compatible with IBD or infectious colitis. Clinical correlation required.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:

A total of 1 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 1 (100.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

The presence of MEDIUM RISK polyps warrants clinical review and correlation with endoscopic findings.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.